

QUALITY TOOLS QUICK REFERENCE

FMEA | SPC | MSA | 8D | APQP | PPAP | IATF 16949 | QMOS Knowledge Base

FMEA - Failure Mode & Effects Analysis

- o Identify failure modes before reaching the customer
- o $RPN = \text{Severity} \times \text{Occurrence} \times \text{Detection}$
- o AIAG-VDA 2019: Action Priority (AP) replaces old RPN ranking
- o PFMEA must link to Control Plan and Work Instructions

SPC - Statistical Process Control

- o Use control charts to detect process instability early
- o $Cp/Cpk \geq 1.67$ for special chars, ≥ 1.33 for others
- o 8 Western Electric rules for out-of-control signals
- o Reaction plan mandatory when OOC signal detected

8D - 8 Disciplines Problem Solving

- o D3 containment protects customer BEFORE root cause is known
- o D4 root cause: why occurred AND why escaped
- o D7 prevention must update FMEA, CP, WI, poka-yoke
- o D8 close only after 90-day effectiveness verification